

BENTONVILLE, AR, USA

WMO: 723444

Lat: 36.350N

Lon: 94.217W

Elev: 395

StdP: 96.67

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03978

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-8.9	-17.5	0.8	-10.2	-14.9	1.1	-7.0	10.1	8.1	9.1	8.5	2.9	310	0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	35.2	23.9	33.2	24.0	32.4	23.6	25.4	32.2	24.8	31.4	24.2	30.7	3.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.0	18.7	28.9	22.7	18.3	28.5	22.4	17.9	28.2	79.8	31.9	77.5	32.0	75.3	30.4	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.9	7.1	DB	-15.7	36.3	2.6	1.6	-17.6	37.4	-19.1	38.3	-20.6	39.2	-22.5	40.3	(4)
(5)			WB	-16.4	26.2	2.5	0.7	-18.2	26.7	-19.7	27.1	-21.1	27.5	-22.9	28.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	2.3	4.6	9.5	14.3	19.0	23.8	26.0	25.6	21.4	15.2	9.2	4.0	(6)
(7)		DBStd	9.53	6.20	6.09	5.71	4.79	4.15	3.00	2.62	2.93	3.95	4.91	5.17	5.57	(7)
(8)		HDD10.0	797	248	168	78	16	2	0	0	0	0	11	75	198	(8)
(9)		HDD18.3	2196	497	384	278	136	43	2	0	1	17	121	275	444	(9)
(10)		CDD10.0	2488	10	18	62	144	281	414	495	483	343	174	52	13	(10)
(11)		CDD18.3	846	0	0	3	14	64	166	237	225	109	25	2	0	(11)
(12)		CDH23.3	8025	0	1	20	102	483	1527	2454	2276	959	193	8	0	(12)
(13)		CDH26.7	3256	0	0	1	9	105	569	1113	1073	347	38	1	0	(13)
(14)	Wind	WSAvg	3.1	3.4	3.6	3.8	3.8	3.1	2.8	2.4	2.5	2.9	3.3	3.3	(14)	
(15)	Precipitation	PrecAvg	1154	64	57	92	113	150	119	97	70	108	84	108	79	(15)
(16)		PrecMax	1664	182	151	197	317	300	312	268	154	263	177	247	325	(16)
(17)		PrecMin	824	4	8	20	39	37	27	24	0	13	26	14	8	(17)
(18)		PrecStd	205	45	39	42	64	57	72	55	37	58	37	69	62	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.1	22.1	26.1	27.7	31.2	34.9	37.2	38.0	33.8	29.7	24.1	20.2	(19)
(20)			MCWB	14.2	14.7	17.0	18.0	21.8	23.8	23.8	23.7	22.1	21.2	16.6	15.1	(20)
(21)		2%	DB	17.5	18.9	22.9	26.2	29.0	32.7	34.8	36.1	32.2	27.6	22.1	17.8	(21)
(22)			MCWB	12.6	12.8	15.2	17.5	21.2	23.8	24.3	23.6	22.4	19.7	15.2	13.4	(22)
(23)		5%	DB	14.9	17.2	21.1	24.1	27.6	31.4	33.0	33.8	30.9	25.8	19.8	15.9	(23)
(24)			MCWB	10.5	11.9	14.1	16.7	20.6	23.4	24.5	23.5	22.0	18.3	14.6	11.7	(24)
(25)		10%	DB	12.4	14.8	18.7	22.6	26.3	30.0	32.3	32.3	28.6	23.0	17.7	12.7	(25)
(26)			MCWB	8.3	10.0	13.2	15.9	19.8	22.9	24.2	23.3	21.2	17.4	13.0	9.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	16.4	18.2	20.5	23.8	25.5	26.1	26.0	24.3	22.6	18.7	16.3	(27)
(28)			MCDWB	18.6	20.0	24.3	25.6	28.7	32.0	32.5	32.7	30.9	27.6	23.4	19.3	(28)
(29)		2%	WB	13.7	14.7	16.6	19.3	22.5	24.7	25.6	25.2	23.6	21.2	17.0	14.4	(29)
(30)			MCDWB	16.5	17.8	21.5	24.0	27.6	30.9	32.2	32.3	29.7	25.8	20.1	17.3	(30)
(31)		5%	WB	11.3	12.6	15.2	18.2	21.6	24.0	25.0	24.5	22.8	19.9	15.4	12.2	(31)
(32)			MCDWB	14.1	16.0	19.4	22.5	26.4	29.9	31.7	31.6	28.4	23.6	18.7	14.8	(32)
(33)		10%	WB	8.3	10.3	13.7	17.0	20.5	23.3	24.4	23.9	21.9	18.6	13.8	9.9	(33)
(34)			MCDWB	11.7	14.3	18.1	21.4	25.3	28.9	30.8	30.7	26.7	22.1	17.2	12.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.5	10.8	11.4	11.8	10.6	10.2	10.6	11.3	11.4	11.8	11.2	10.0	(35)
(36)			MCDBR	13.5	13.9	14.3	13.3	11.5	11.6	11.8	13.6	12.8	13.4	13.0	13.2	(36)
(37)			MCWBR	9.3	9.5	8.6	7.5	6.0	5.1	4.5	4.7	5.2	7.0	8.3	9.3	(37)
(38)		5% WB	MCDBR	11.3	11.8	11.9	10.7	10.3	10.4	10.8	11.3	10.7	10.5	11.3	11.0	(38)
(39)			MCWBR	9.1	9.5	8.3	7.1	6.0	5.3	4.7	4.5	4.9	6.3	8.6	9.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.302	0.305	0.339	0.376	0.420	0.421	0.442	0.443	0.386	0.343	0.311	0.291	(40)	
(41)		taud	2.484	2.485	2.423	2.328	2.259	2.280	2.275	2.249	2.378	2.482	2.520	2.534	(41)	
(42)		Ebn at Noon	897	938	929	906	867	862	841	833	868	881	880	890	(42)	
(43)		Edh at Noon	82	92	108	125	136	134	133	134	112	92	79	74	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.45	2.99	4.02	5.08	5.63	6.44	6.37	5.78	4.94	3.67	2.64	2.09	(44)	
(45)		RadStd	0.22	0.42	0.24	0.29	0.47	0.56	0.39	0.40	0.38	0.47	0.29	0.26	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page